Unit: Unit 0: Foundations of Algebra
Topic: Rational Numbers Decimals

Name: ________________
Date: ____________

Foundation (Jalapeno)

Q1. What is the decimal equivalent of the fraction $\frac{3}{4}$?
(A) 0.25
(B) 0.5
(C) 0.75
(D) 0.8
(E) 0.9

Q2. A basketball player scored $\frac{7}{10}$ of the team's points. What percent of the points did the player score?
(A) 50
(B) 60
(C) 65
(D) 70
(E) 75

Q3. Which of the following rational numbers is greatest: $\frac{2}{3}$, $\frac{3}{4}$, $\frac{5}{6}$?
(A) $\frac{2}{3}$
(B) $\frac{3}{4}$
(C) $\frac{5}{6}$
(D) $\frac{2}{3}$ and $\frac{3}{4}$ are equal
(E) None of the above

Q4. A cyclist traveled 2.5 miles in $\frac{1}{2}$ hour. How many miles did the cyclist travel per hour?
(A) 4 miles
(B) 5 miles
(C) 6 miles
(D) 7 miles
(E) 8 miles

Q5. What is the fraction equivalent of the decimal 0.6?
(A) $\frac{1}{10}$
(B) $\frac{1}{5}$
(C) $\frac{3}{5}$
(D) $\frac{2}{3}$
(E) $\frac{3}{4}$

Q6. A tennis player won $\frac{4}{5}$ of her matches. What percent of her matches did she win?
(A) 60
(B) 70
(C) 75
(D) 80
(E) 85

Q7. Which of the following rational numbers is least: $\frac{1}{2}$, $\frac{1}{3}$, $\frac{2}{5}$?
(A) $\frac{1}{2}$
(B) $\frac{1}{3}$
(C) $\frac{2}{5}$
(D) $\frac{1}{2}$ and $\frac{1}{3}$ are equal
(E) None of the above

Q8. A swimmer finished a lap in 1.8 minutes. What fraction of an hour did the swimmer take to finish the lap?
(A) $\frac{1}{20}$
(B) $\frac{1}{30}$
(C) $\frac{1}{50}$
(D) $\frac{3}{100}$
(E) $\frac{1}{100}$

Open Ended Questions (FRQ)

Q9. Represent the rational numbers $\frac{1}{4}$, $\frac{1}{2}$, and $\frac{3}{4}$ on a number line and explain your reasoning.

Q10. A gymnast scored 85 points out of 100. Express this score as a fraction and a decimal. Calculate the percentage score and show your work.

Chef's Tips

- Ensure students understand the concept of rational numbers and their representation on a number line.
- Encourage students to show their work and explain their reasoning for FRQs.
- Remind students to convert between fractions, decimals, and percents accurately.